18.3. $A = 10.0\,cm^2 = 1.0\times10^{-3}\,m^2$

$T = 2900\,K$

$\varepsilon = 1$, $b = 2.898\times10^{-3}\,m\cdot K$. $\sigma = 5.67\times10^{-8}\,W/(m^2\cdot K^4)$

(1) $\lambda_m = \dfrac{b}{T} = \dfrac{2.898\times10^{-3}\,m\cdot K}{2900\,K} \approx 1000\,nm$

(2) $P = A\cdot\sigma T^4$

$= 1.0\times10^{-3}\times5.67\times10^{-8}\times2900^4$

$\approx 4010\,W$

18.7 $\theta = \dfrac{\pi}{2}$

(1) $\Delta\lambda = \lambda_C = 0.002426\,nm$

$\lambda' = \lambda + \Delta\lambda = 0.0708 + 0.002426 \approx 0.0732\,nm$

$E_K = hc\left(\dfrac{1}{\lambda} - \dfrac{1}{\lambda'}\right) = 1240\times\left(\dfrac{1}{0.0708} - \dfrac{1}{0.073226}\right) \approx 0.580\,keV$

(2) $\theta = \pi$

$\Delta\lambda = 2\lambda_C = 0.004852\,nm$

$\lambda' = \lambda + \Delta\lambda \approx 0.757\,nm$

$E_C = hc\left(\dfrac{1}{\lambda} - \dfrac{1}{\lambda'}\right) \approx 1.12\,keV.$

18.13

$E_K = qU$

$P = \sqrt{2mE_K} = \sqrt{2mqU}$

$\lambda = \dfrac{h}{P} = \dfrac{h}{\sqrt{2mqU}}$

$m = \dfrac{h^2}{2qU\lambda^2} = \dfrac{(6.63\times10^{-34})^2}{2\times1.6\times10^{-19}\times206\times(2.0\times10^{-12})^2} \approx 1.67\times10^{-27}\,kg$

18.24. $E = \dfrac{p^2}{2m} + \dfrac{1}{2}m\omega^2x^2$

$x^2 \approx (\Delta x)^2$, $p^2 \approx (\Delta p)^2$

$E \approx \dfrac{(\Delta p)^2}{2m} + \dfrac{1}{2}m\omega^2(\Delta x)^2$

$\Delta p \approx \dfrac{\hbar}{2\Delta x}$

$E(\Delta x) \approx \dfrac{\hbar^2}{8m(\Delta x)^2} + \dfrac{1}{2}m\omega^2(\Delta x)^2$

$\dfrac{dE}{d(\Delta x)} = -\dfrac{\hbar^2}{4m(\Delta x)^3} + m\omega^2(\Delta x) = 0$

$(\Delta x)^2 = \dfrac{\hbar}{2m\omega}$

$E_{min} \approx \dfrac{\hbar^2}{8m\cdot\frac{\hbar}{2m\omega}} + \dfrac{1}{2}m\omega^2\cdot\dfrac{\hbar}{2m\omega} = \dfrac{1}{2}\hbar\omega$